AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2017

Biology Paper II

Time: 2 hours 20 minutes Marks: 55

AKU-EB May Examination 2017

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are ELEVEN questions. Answer ALL questions. Questions 10 and 11 offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

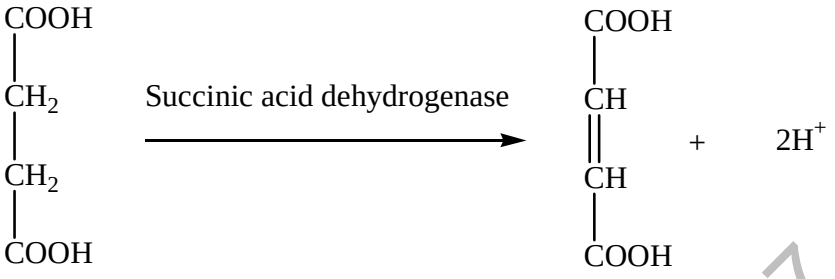
   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 4 Marks)

a. Consider the given reactions **I** and **II**.

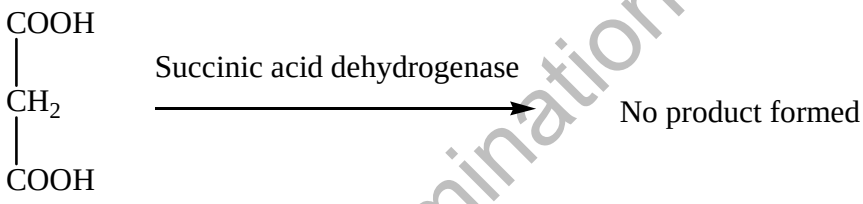
Reaction I



Succinic acid

Fumaric acid

Reaction II



Malonic acid

If succinic acid is added with malonic acid in reaction **II** and its quantity is 10 times that of malonic acid, will a product be formed? Justify your answer. (2 Marks)

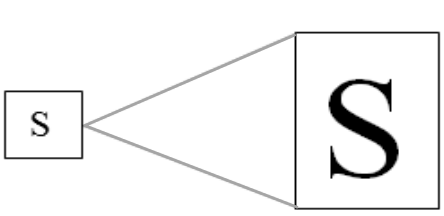
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b. The given diagrams show two features of the microscope. Identify these features. (2 Marks)



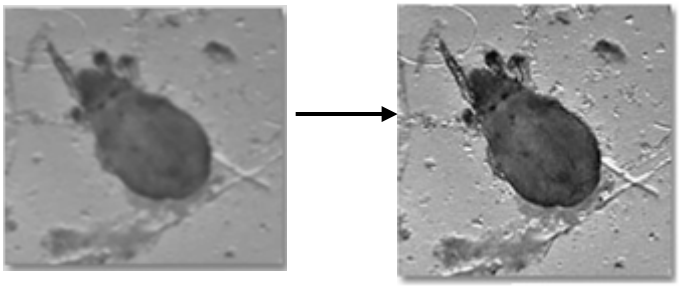


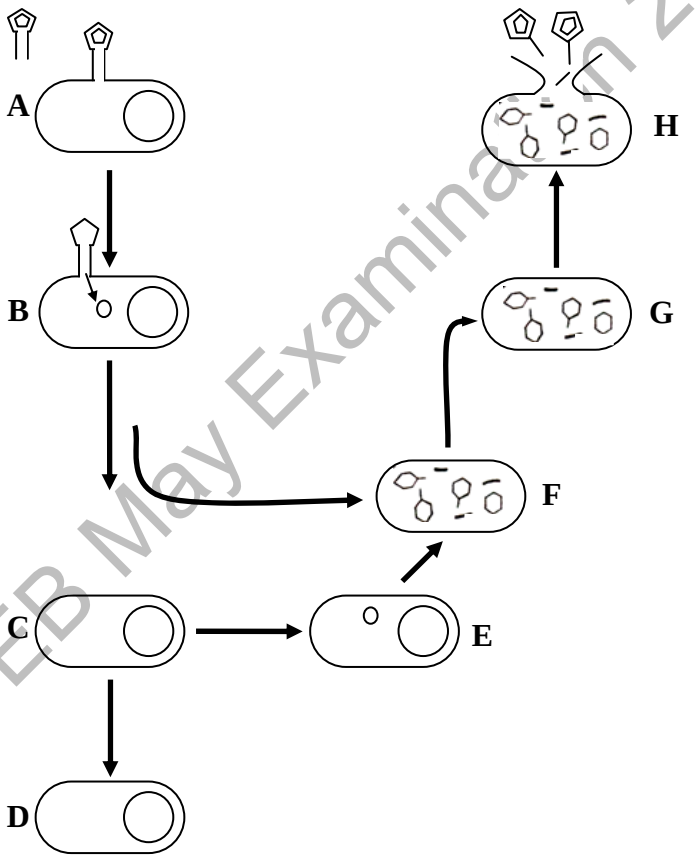
Diagram I: _____ **Diagram II:** _____

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Q.2. (Total 4 Marks)

a. Describe the use of bacteriophages in genetic engineering. (2 Marks)

b. The given diagram shows the life cycle of a bacteriophage.



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i. Which common action is taking place at stage **B** and stage **H**? (1 Mark)

ii. At which stage does the process of induction take place? (1 Mark)

Q.3. (Total 4 Marks)

a. Describe magnetotaxis in bacteria. (2 Marks)

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b. State any TWO roles of mesosomes in a bacterial cell. (2 Marks)

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Q.4. (Total 4 Marks)

a. List any TWO common characteristics of actinopods and amoebae. (2 Marks)

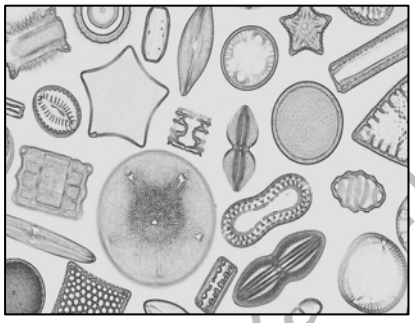
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b. The given diagram shows some organisms of the kingdom Protista.



i. What is the composition of shells of the organisms shown in the diagram? (1 Mark)

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ii. Mention the importance of these organisms to the ecosystem. (1 Mark)

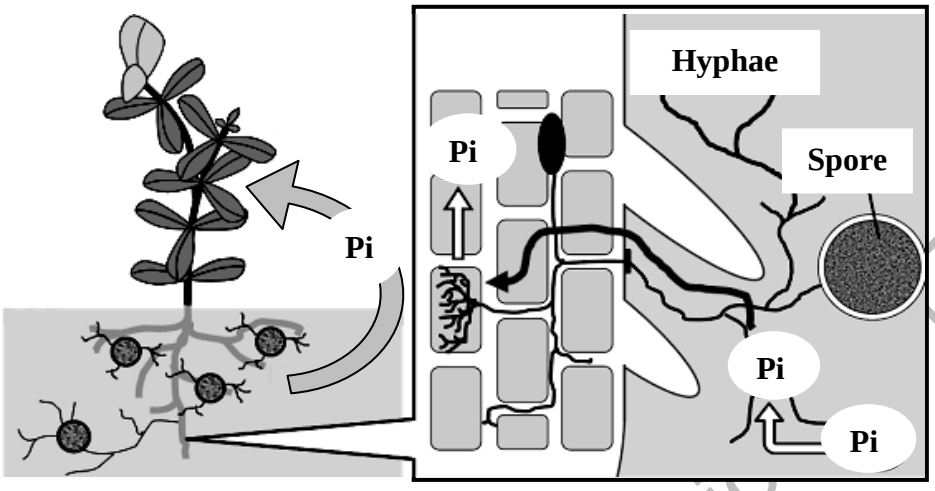
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Q.5. (Total 4 Marks)

The given diagram shows a mycorrhizal association.



i. Identify the type of mycorrhizal association. Give a reason to support your answer. (2 Marks)

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_____

_____

_____

ii. How do the fungal hyphae increase the absorption of inorganic phosphate (Pi) in plants? (1 Mark)

_____

_____

iii. What is the benefit of mycorrhizal association to fungi? (1 Mark)

_____

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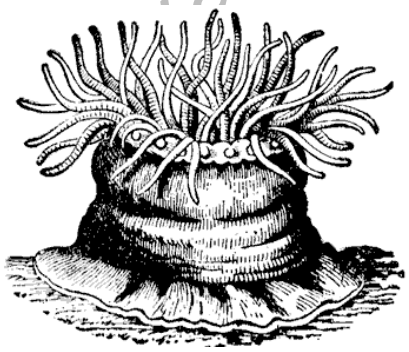
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Q.6. (Total 5 Marks)

a. Fill in the given table by stating the names of part of gastrointestinal tract which match each mentioned pH level and also state the role of pH on enzymatic digestion. (4 Marks)

pH	Name of the Part of Gastrointestinal Tract	Role of pH in Enzymatic Digestion
High (around 8)		
Low (around 2)		

b. The given diagram shows the radial symmetry of an invertebrate.



In which body form does this invertebrate exist? (1 Mark)

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Q.7. (Total 5 Marks)

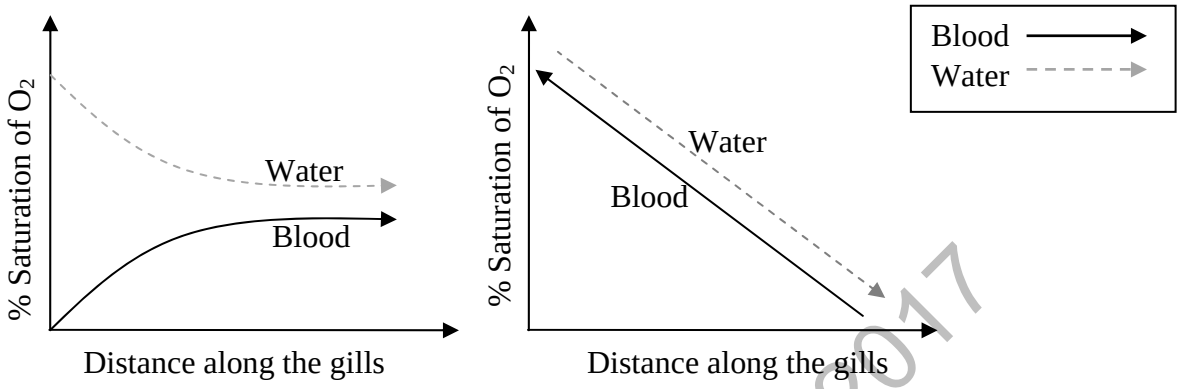
a. There are various steps that took place in the evolution of seed habit, out of which, one is 'reduction to a single functional megaspore per sporangium'. Describe the need and outcome of this step. (3 Marks)

b. Describe the condition that leads to the shift in the flow of electrons from non-cyclic to cyclic phosphorylation during the light dependent reactions of photosynthesis. (2 Marks)

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Q.8. (Total 5 Marks)

a. The given graphs show two mechanisms of exchange of gases.



i. Which graph represents the mechanism of exchange of gases in fish? (1 Mark)

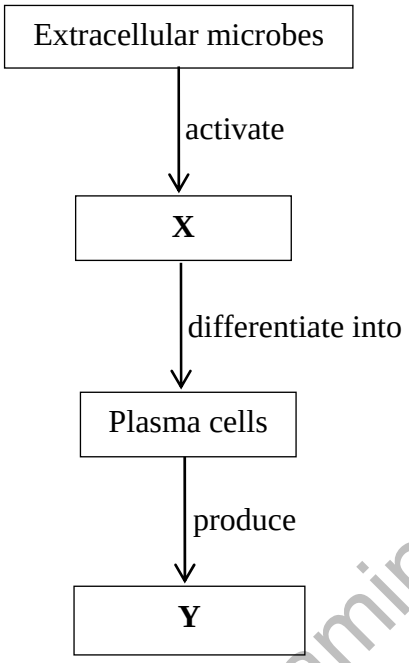
ii. 'The exchange of gases is efficient in fish.' Compare both the graphs to explain this statement. (2 Marks)

b. Mention any TWO similarities between lymphatic vessels and veins? (2 Marks)

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Q.9. (Total 5 Marks)

a. The given flow chart represents the immune response produced upon the invasion of extracellular microbes in the human body.



Identify **X** and **Y** in the given flowchart. (2 Marks)

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b. Identify the phenomena that are described in the given table with reference to the ascent of sap in plants. (3 Marks)

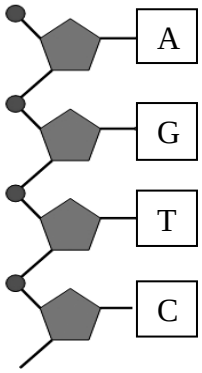
Description of Phenomenon	Name of Phenomenon
Water molecules stick to the walls of xylem so that the column of water in xylem tissue does not break.	
It is the uptake of water by seeds during germination.	
It is provided when the water chain is pulled up in the xylem. It is formed between the molecules of water by hydrogen bonding.	

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Q.10 (Total 7 Marks)

EITHER

a. The given diagram depicts a strand of DNA.



i. Using the given strand as template, illustrate the complementary strand to complete the double helical structure of DNA. (2 Marks)

ii. Describe any FIVE properties of double helical structure of DNA. (5 Marks)

OR

b.

i. Describe the function of ribosomes, endoplasmic reticulum, Golgi apparatus and plasma membrane in the formation and secretion of proteins. (4 Marks)

ii. Describe the role of endoplasmic reticulum and Golgi apparatus in the formation of lysosomes. (3 Marks)

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Q.11. (Total 8 Marks)

EITHER

a. According to an article published by Nic Fleming on 10 February, 2015, "The most dominating form of life, in terms of their number on the Earth, is six-legged, shrimp-like springtails … The 6,000 known species of these wingless arthropods can be found in all manner of habitats all over the world, from beaches and cliffs to the Antarctic and the highest mountain ranges on the Earth."

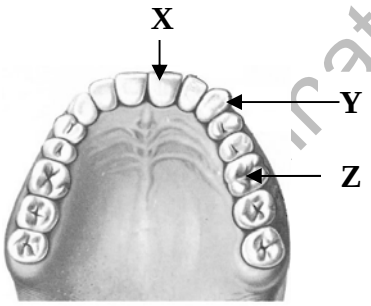
i. Explain any FOUR ways in which the organisms described in the given text have successfully adapted to diverse habitats. (4 Marks)

ii. Describe any FOUR ways in which these organisms impact our economy. (4 Marks)

OR

b.

i. The given diagram shows different types of human teeth. (3 Marks)



Write the function of teeth labelled as **X**, **Y** and **Z** in the given diagram.

ii. Describe any FIVE steps involved in the movement of bolus from oral cavity to the stomach. (5 Marks)

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